

Mitochondrial Genome Instability and mtDNA Depletion in Human Cancers

HSIN-CHEN LEE,^a PEN-HUI YIN,^a JIN-CHING LIN,^b CHENG-CHUNG WU,^c
CHIH-YI CHEN,^d CHEW-WUN WU,^e CHIN-WEN CHI,^{a,f} TSENG-NIP TAM,^g
AND YAU-HUEI WEI^h

*Departments of Pharmacology^a and Biochemistry,^h School of Medicine,
National Yang-Ming University, Taiwan 112, Republic of China*

*^bDepartment of Radiation Oncology, Section of General Surgery^c and
of Thoracic Surgery,^d Department of Surgery, Taichung Veterans General Hospital,
Taiwan 407, Republic of China*

*Departments of Surgery^e and Medical Research and Education,^f
Taipei Veterans General Hospital, Taiwan 112, Republic of China*

*^gSection of Gastroenterology, Department of Internal Medicine,
Cheng Hsin Rehabilitation Medical Center, Taipei, Taiwan 112, Republic of China*

ABSTRACT: An increase in the rate of glycolysis is one of the metabolic alterations in most cancer cells. However, the role of alterations in mitochondrial function and mitochondrial DNA (mtDNA) in carcinogenesis still remains unclear. In this study, we analyzed the nucleotide sequence of the D-loop and the copy number of mtDNA in 54 hepatocellular carcinomas (HCCs), 31 gastric, 31 lung, and 25 colorectal cancers as well as their corresponding non-tumorous tissues. The results revealed that 42.6% (23/54) of the HCCs, 51.6% (16/31) of the gastric cancers, 22.6% (7/31) of the lung cancers, and 40.0% (10/25) of the colorectal cancers harbored mutation(s) in the D-loop of mtDNA. The mtDNA mutations in 43.5% (10/23) of the HCCs, 62.5% (10/16) of the gastric cancers, 57.1% (4/7) of the lung cancers, and 90.0% (9/10) of the colorectal cancers were changes in the mononucleotide or dinucleotide repeats, deletions, or multiple insertions. Moreover, we found that there is a significant decrease in mtDNA copy number in 57.4% (31/54) of the HCCs, 54.8% (17/31) of the gastric cancers, 22.6% (7/31) of the lung cancers, and 28.0% (7/25) of the colorectal cancers compared with the corresponding non-tumorous tissues. It is noteworthy that the incidence of somatic mutations in the D-loop of mtDNA in the cancers of later stages was higher than that of the early-stage cancers. Taken together, our findings suggest that instability in the D-loop region of mtDNA, together with the decrease in mtDNA copy number, is involved in the carcinogenesis of human cancers.

KEYWORDS: hepatocellular carcinoma; gastric cancer; lung cancer; colorectal cancer; mitochondrial DNA; somatic mutation; copy number

Address for correspondence: Dr. Hsin-Chen Lee, Department of Pharmacology, School of Medicine, National Yang-Ming University, Taipei, Taiwan 112, Republic of China. Voice: +886-2-28267327; fax: +886-2-28264372.
hclee2@ym.edu.tw

Ann. N.Y. Acad. Sci. 1042: 109–122 (2005). © 2005 New York Academy of Sciences.
doi: 10.1196/annals.1338.011

INTRODUCTION

An increase in the capacity of glycolysis accompanied by defective respiration is one of the common alterations in bioenergetic function in many cancer cells. As early as 75 years ago, Warburg hypothesized that cancer cells may have impaired mitochondrial function and that this alteration would result in an elevated rate of glycolysis.¹ In the past few years, it has been demonstrated that glycolysis could be activated by hypoxia and modulated by alteration in the expression of oncogenes or tumor suppressor genes.² However, the presumed impairment of the bioenergetic function of mitochondria has never been established in human cancers.

Mitochondria are responsible for the supply of most of the energy needed by the human cell, and they play a key role in the initiation and execution of apoptosis.³ Moreover, mitochondria are the major intracellular producer of reactive oxygen species (ROS) and are also subject to direct attack by ROS in the organelles of mammalian cells.³ Impairment in mitochondrial respiratory function not only reduces the supply of energy, which may prevent energy-dependent apoptosis, but also enhances ROS production that may induce mutation and oxidative damage to mitochondrial DNA (mtDNA). It has been proposed that accumulation of mtDNA mutations and alteration in the execution of apoptosis contribute to the onset and progression of various neurodegenerative diseases.³ Recently, it was reported that the expression level of the β subunit of F₁-ATPase required for mitochondrial ATP synthesis is decreased in human cancers, including liver, gastric, lung, and colorectal cancers.^{4,5} These findings suggest that there are common mechanisms by which the bioenergetic function of mitochondria is altered in cancer cells. However, little information is available as to how qualitative and quantitative changes in mtDNA are involved in the alteration of mitochondrial oxidative phosphorylation in cancer tissues.

Human mtDNA is a 16.5-kb circular double-stranded DNA molecule, which is present at a high copy number per cell; the number varies widely with the cell type. Human mtDNA contains genes coding for 13 polypeptides involved in respiration and oxidative phosphorylation, 2 rRNAs, and a set of 22 tRNAs that are essential for the protein synthesis in mitochondria.⁶ In addition, mtDNA contains a noncoding region that includes a unique displacement loop (D-loop), which controls replication and transcription of mtDNA.⁶ It has been established that the mitochondrial genome is particularly susceptible to oxidative damage and mutation because of the high rate of ROS generation and inefficient DNA repair system in mitochondria.^{7,8} Both inherited and somatic mtDNA mutations have been demonstrated to cause severe degenerative disorders.^{8,9} Most mutations occur in the tRNA genes and structural genes of mtDNA in the affected tissues of patients with inherited mitochondrial disorders. By contrast, many of the mtDNA mutations found in human cancers are located in the D-loop region, which has been shown to be a "hot spot" for point mutations in many human cancers.¹⁰⁻¹⁴ Because the D-loop is involved in the control of replication and transcription of mtDNA, mutations in this region might cause a decrease in the copy number and/or alteration in gene expression of the mitochondrial genome.¹⁵

To investigate qualitative and quantitative alterations in mtDNA of human cancer, we analyzed the nucleotide sequence of the D-loop and determined the copy number of mtDNA in the cancerous tissues and corresponding non-cancerous tissues from patients with hepatocellular carcinoma (HCC), gastric cancers, lung cancers,

and colorectal cancers. The consequence of mtDNA mutations and the mechanisms by which the copy number of mtDNA is changed in human cancers are discussed.

MATERIALS AND METHODS

Collection of Human Cancer Tissues and DNA Extraction

Fifty-four HCCs, 31 lung cancers, and corresponding non-tumorous tissues were obtained with consent from patients and were histologically confirmed at Taichung Veterans General Hospital. Thirty-one gastric cancers and corresponding non-tumorous tissues were obtained with consent from patients at Taipei Veterans General Hospital. In addition, 25 colorectal cancers and corresponding non-tumorous tissues were obtained with consent from patients at Cheng Hsin Rehabilitation Medical Center in Taipei. All the tissues were kept in liquid nitrogen immediately after surgical resection according to a protocol approved by the medical ethics committee for conducting human research at the three hospitals. Total DNA of these tissues was extracted by proteinase K/SDS lysis followed by phenol/chloroform extraction as previously described.¹⁶ The DNA was dissolved in doubly distilled water and frozen at -30°C until use.

Direct Sequencing for Determination of Somatic Mutation in the mtDNA D-Loop

The somatic mutation in the D-loop region of mtDNA was analyzed by direct sequencing of the PCR products. The primer pairs L16190 (nucleotide position [np] 16190–16209, 5'-CCCCATGCTTACAAGCAAGT-3') and H602 (np 602–583, 5'-GCTTTGAGGAGGTAAGCTAC-3') were used for the amplification of a 982-bp DNA fragment from the D-loop region of mtDNA. All polymerase chain reaction (PCR) products were purified and sequenced with the AmpliCycle sequencing kit (Perkin-Elmer/Cetus) according to the instructions of the manufacturer.

Determination of mtDNA Copy Number

A competitive PCR method was used for determining the copy number of mtDNA.^{16,17} In brief, a known amount of the internal DNA standard (a competitor DNA fragment that contained truncated ND1 and β -actin gene) was added with the DNA sample into the PCR reaction mixture. The PCR was carried out for 25 cycles in a 50-mL reaction mixture containing 200 ng DNA, 200 μM of dNTP, 40 pmol of each primer, 1.0 U of *Taq* DNA polymerase, 50 mM KCl, 1.5 mM MgCl_2 , 10 mM Tris-HCl (pH 9.0), 0.1% Triton X-100, and 0.01% (w/v) gelatin. The sequences of the primers are β -actin forward primer: 5'-CATGTGCAAGGCCGGCTTCG-3'; β -actin reverse primer: 5'-CTGGGTCATCTTCTCGCGGT-3'; ND1 L-strand primer: 5'-TCTCACCATCGCTCTTCTAC-3'; and ND1 H-strand primer: 5'-TTG-GTCTCTGCTAGTGTGGA-3'. The PCR cycles consisted of denaturation at 94°C for 15 s, annealing at 58°C for 15 s, and primer extension at 72°C for 40 s. The PCR products were separated by electrophoresis on a 3% agarose gel at 100 V for 40 min and detected under UV transillumination after ethidium bromide staining. The band intensities of the PCR products of the target and internal standard DNAs were ana-

TABLE 1. Somatic mutations in the D-loop region of mtDNA of human cancers

Nucleotide position	Type of mutation	HCC	Gastric cancer	Lung cancer	Colorectal cancer	Homoplasmy
16260	C→T		1			Yes
16298	C→T	1 ^a				Yes
16300	A→G/T	1				heteroplasmy
16390	G→A/G			1		heteroplasmy
16391	G→A/G			1		heteroplasmy
16399	A→A/G		1			heteroplasmy
16438	G→G/A		1 ^d			heteroplasmy
16519	C→T	1	1 ^d			Yes
70	G→A	1				Yes
72	T→C	4 + 1 ^b				Yes
94	G→A	1				Yes
146	T→C	1				Yes
152	T→C	1 ^a				Yes
189	A→G	1	1		1 ^e	Yes
200	G→A				1	Yes
204	T/C→T/C		1			heteroplasmy
205	G→A		1			Yes
303	7C→8C	1				heteroplasmy
303	8C→7C	2	2			Yes
303	8C→7C		1	1		heteroplasmy
303	8C→9C	1	1		3	Yes
303	8C→9C	1	4	1		heteroplasmy
303	9C→7C	1			1	Yes
303	9C→8C	1			4	Yes
303	9C→8C	1	1 + 1 ^d	1		heteroplasmy
303	9C→10C				1 ^e	Yes
303	9C→10C			1		heteroplasmy
303	9C→13C	1 ^b				heteroplasmy
303	9C→14C	1				heteroplasmy
417	G→A	1 ^c				Yes
514	5CA→4CA	1 ^c				Yes
298–348/306–356	deletion	1		1		heteroplasmy
311/568	insertions		1			heteroplasmy

^{a, b, c, d e}: The indicated mtDNA mutations were detected in the same tumor samples with identical superscripts.

lyzed. The ratio between the intensities of the two DNA bands was used to calculate the relative DNA content of the target gene. The DNA content of the ND1 gene was normalized with that of the β -actin gene in nuclear DNA to calculate the copy number of mtDNA in each sample.

RESULTS

Somatic Mutation in the D-Loop of mtDNA of Human Cancers

To search for tumor-associated somatic mutation in the D-loop of mtDNA, we directly analyzed the nucleotide sequences in the D-loop region of mtDNA in tumors and in corresponding non-tumorous tissues. The results showed that 42.6% (23/54)

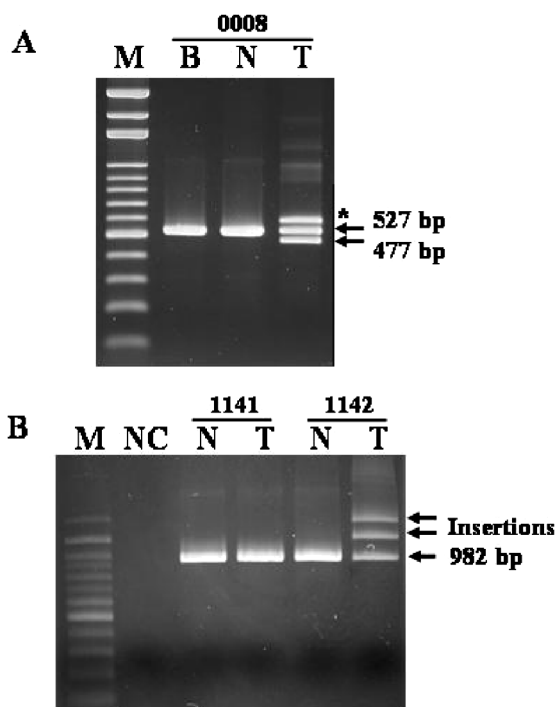


FIGURE 1. The deletion and insertion of mtDNA in human cancers. Human mtDNA with a deletion or insertion in the D-loop was detected by the PCR technique described in MATERIALS AND METHODS. Using the primers L76 (5'-CACGCGAATAGCATTGC-GAGACGCTG-3') and H602, the 527-bp and 477-bp PCR fragments amplified from the wild-type mtDNA and the 50 bp-deleted mtDNA, respectively, in the tumor part (T) and in the surrounding non-tumor part (N) of lung cancer patient 0008 (A) were separated on a 1.5% agarose gel and detected under UV transillumination after ethidium bromide staining. M, 100-bp DNA ladder; B, blood sample; *, one insertion in the D-loop region of mtDNA in the case. Multiple insertions in the D-loop region of mtDNA in the gastric cancer of patient 1142 (B) were detected by using the primers L16190 and H602. NC, negative control.

of the HCCs, 51.6% (16/31) of the gastric cancers, 22.6% (7/31) of the lung cancers, and 40.0% (10/25) of the colorectal cancers harbored somatic mutation(s) in the D-loop region of mtDNA (TABLE 1). Moreover, most of the mtDNA mutations in HCCs and colorectal cancers were homoplasmic, but those in the gastric cancers and lung cancers were mostly heteroplasmic. Interestingly, 43.5% (10/23) of the mtDNA mutations detected in the HCCs, 62.5% (10/16) in the gastric cancers, 57.1% (4/7) in the lung cancers, and 90.0% (9/10) in the colorectal cancers were alterations in the mononucleotide repeat located in the polycytidine stretch between np 303–309 of mtDNA (TABLE 1). Furthermore, one HCC contained a change in the dinucleotide repeat (CA)_n at np 514; and two cancers (one HCC and one lung cancer) carried a specific 50-bp deletion (FIG. 1A), which was located between np 298/306 and 348/356, and the breakpoints were flanked by a 9-bp direct repeat of 5'-CCAAACCCC-3'.¹⁷ A gastric cancer sample harbored multiple insertions located between np 311 and np 568 of mtDNA (FIG. 1B).

Depletion of mtDNA in Human Cancers

Using a competitive PCR method, we determined the mtDNA copy number of tumors and of corresponding non-tumorous tissues. The results revealed that 57.4% (31/54) of the HCCs, 54.8% (17/31) of the gastric cancers, 22.6% (7/31) of the lung

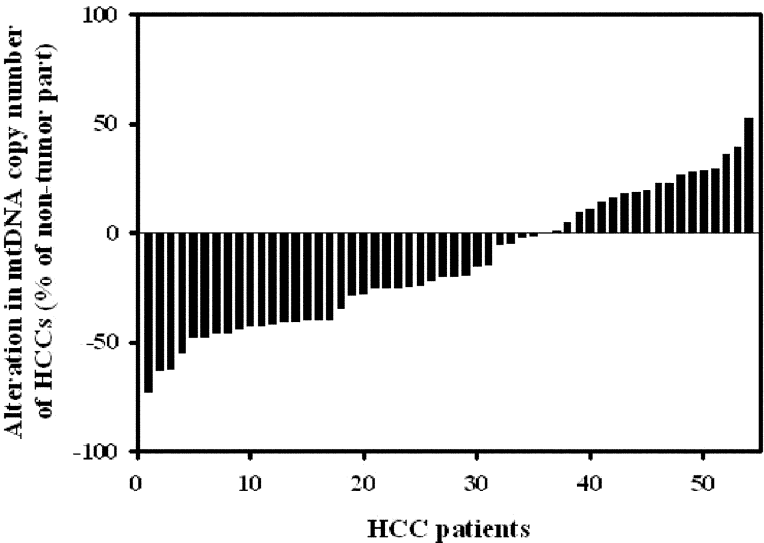


FIGURE 2. Alteration of the copy number of mtDNA in hepatocellular carcinomas. The copy numbers of mtDNA in 54 HCCs and their corresponding non-tumorous tissues were determined by a competitive PCR method described in MATERIALS AND METHODS. The alteration of the copy number of mtDNA in each HCC sample was indicated as the relative percentage of the copy number of mtDNA in the corresponding non-tumorous tissue. A negative value represents a decrease in the copy number of mtDNA in the cancer, while a positive value represents an increase in the mtDNA copy number.

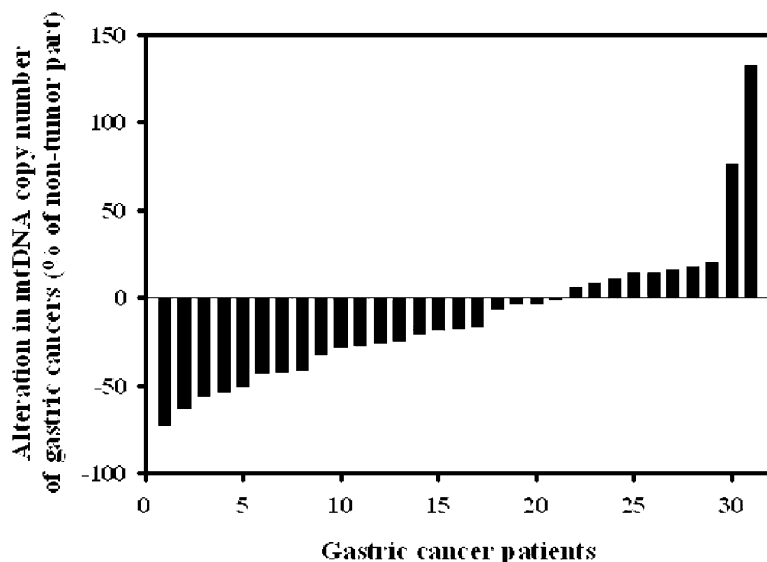


FIGURE 3. Alteration of the copy number of mtDNA in gastric cancers. The copy numbers of mtDNA in 31 gastric carcinomas and their corresponding non-tumorous tissues were determined by a competitive PCR method described in MATERIALS AND METHODS. An alteration of the copy number of mtDNA in each gastric cancer sample was indicated as the relative percentage of the copy number of mtDNA in the corresponding non-tumorous part of the tissue. A negative value represents a decrease in the copy number of mtDNA in the cancer, while a positive value represents an increase in the mtDNA copy number.

cancers, and 28.0% (7/25) of the colorectal cancers had significant decreased mtDNA copy number (below 90%) as compared with the corresponding non-tumorous tissues (FIGS. 2–5). By contrast, 27.8% (15/54) of the HCCs, 22.6% (7/31) of the gastric cancers, 48.4% (15/31) of the lung cancers, and 40.0% (10/25) of the colorectal cancers had significantly higher mtDNA copy numbers (above 110%) as compared with the corresponding non-tumorous tissues (FIGS. 2–5).

Higher Incidence of Mutation in the D-Loop of mtDNA in Cancers at Later Stages

After analysis of the incidence of somatic mutation in the D-loop and depletion of mtDNA in the cancers at various stages, we found that the incidence of somatic mutation of mtDNA in all four kinds of cancers at stages III and IV was higher than those at stages I and II (TABLE 2). In addition, the incidence of mtDNA depletion in colorectal cancers at stages III and IV was higher than those at stages I and II, respectively. However, the increase did not occur in HCCs, gastric cancers, or lung cancers (TABLE 2).

TABLE 2. Incidence of mutation and depletion of mtDNA in human cancers at different tumor stages

TNM stages	Incidence of mtDNA alteration in cancers	
	Somatic mutation in the D-loop (%)	Depletion of mtDNA (%)
HCC		
I + II	9/25 (36)	15/25 (60.0)
III + IV	14/29 (48.3)	16/29 (55.2)
Gastric cancer		
I + II	2/5 (40.0)	3/5 (60.0)
III + IV	14/26 (53.8)	14/26 (53.8)
Lung cancer		
I + II	4/21 (19.0)	5/21 (23.8)
III + IV	3/10 (30.0)	2/10 (20.0)
Colorectal cancer		
I + II	3/11 (27.3)	2/11 (18.2)
III + IV	7/14 (50.0)	5/14 (35.7)

DISCUSSION

Our results revealed that among a total of 141 Taiwanese cancer patients (including 54 HCCs, 31 gastric cancers, 31 lung cancers, and 25 colorectal cancers) examined, 39.7% of the examined cancerous tissues carried mutation(s) in the D-loop of mtDNA. The alterations of mononucleotide repeat located in the polycytidine stretch at np 303–309 were the most frequent ones in these cancers (TABLE 1). The change in the dinucleotide repeat (CA)_n at np 514, the specific 50-bp deletion, and multiple insertions were detected in some cancerous tissues. Moreover, the incidence of somatic mutation in the D-loop of mtDNA in these cancers was found to increase with the cancer stages. These findings have supported the notion that the alterations are a result of mtDNA instability in most cancers during the carcinogenesis process.

It is worth noting that the most frequent changes in mtDNA were located in the polycytidine stretch, as well as the specific 50-bp deletion, and multiple insertions were flanked by the sequences containing polycytidine run in the D-loop region. It was recently demonstrated that the D-loop was highly susceptible to oxidative damage as compared with the other regions of mtDNA.¹⁸ The extensive oxidative damage to the polycytidine sequences may result in slipping and/or misincorporation during replication or repair of mtDNA by mitochondrial DNA polymerase. The DNA polymerase γ is the sole DNA polymerase in mitochondria and is responsible for replication and repair of mtDNA. It has been demonstrated that mitochondrial DNA polymerase γ is a target of oxidative damage.¹⁹ Oxidative stress can significantly inhibit DNA polymerase activity and lower its DNA binding efficiency,

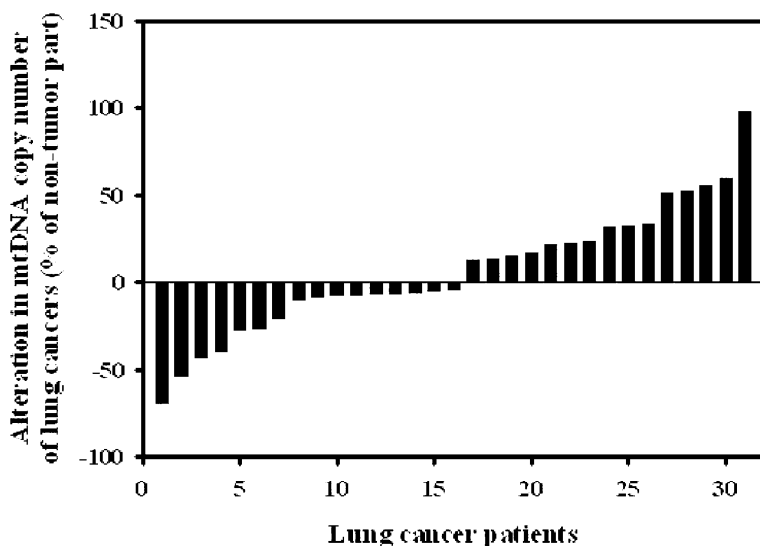


FIGURE 4. Alteration of the copy number of mtDNA in lung cancers. The copy numbers of mtDNA in 31 lung carcinomas and their corresponding non-tumorous tissues were determined by a competitive PCR method described in MATERIALS AND METHODS. An alteration of the copy number of mtDNA in each lung cancer sample was indicated as the relative percentage of the copy number of mtDNA in the corresponding non-tumorous part. The negative value represents the decrease in the copy number of mtDNA in the cancer, while a positive value represents an increase in the copy number of mtDNA.

which may result in less efficient replication and repair of mtDNA.¹⁹ Therefore, oxidative damage to mitochondrial DNA polymerase γ may also contribute to the alteration in the number of polycytidine during replication or repair of mtDNA (FIG. 6).

In addition, we found that 44.0% of the cancers examined in this study had a lower copy number of mtDNA as compared with the corresponding non-tumorous tissues, but 33.3% of the cancers had higher mtDNA content. It was recently reported that point mutation in the D-loop region is associated with a decrease in mtDNA copy number in HCC.¹⁷ Due to the role of the D-loop in controlling both replication and transcription of mtDNA,¹⁵ oxidative damage and/or somatic mutations in the region might interfere with replication and maintenance of mtDNA. Moreover, it was reported that a deregulation in the expression of mitochondrial single-stranded DNA binding protein (mtSSB) is associated with the decrease in mtDNA copy number of HCC.²⁰ Thus, oxidative damage to the D-loop of mtDNA and/or less efficiency in the DNA replication could be involved in the instability and depletion of mtDNA in the cancerous tissues (FIG. 6).

The mitochondrial genome is more susceptible to DNA damage and consequently acquires mutations at a higher rate than does nuclear DNA because of the lack of histone protection of mtDNA and the high rate of ROS generation and limited DNA repair mechanisms in human mitochondria. It is plausible that when the frequency

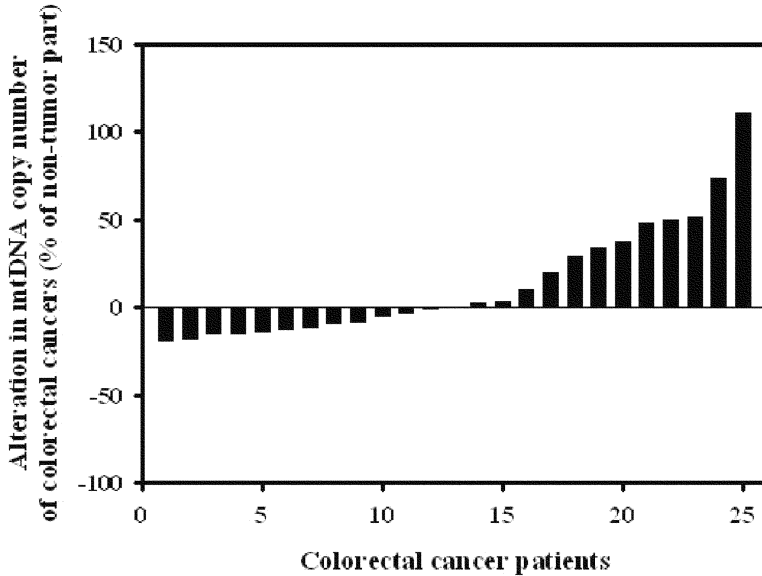


FIGURE 5. Alteration of the copy number of mtDNA in colorectal cancers. The copy numbers of mtDNA in 25 colorectal carcinomas and their corresponding non-tumorous tissues were determined by a competitive PCR method described in MATERIALS AND METHODS. An alteration of the copy number of mtDNA in each colorectal cancer sample was indicated as a relative percentage of the copy number of mtDNA in the corresponding non-tumorous part. A negative value represents the decrease in the copy number of mtDNA in the cancer, while a positive value represents an increase in the mtDNA copy number.

of occurrence and proportion of mtDNA mutations are increased in somatic tissues, nuclear DNA may be subjected to damage or mutation, although at a lower level. Thus, mtDNA mutation and oxidative damage can be used as an index of genome instability in cancer cells. These observations are consistent with the concept that tumor cells are exposed to a higher and persistent oxidative stress compared with the adjacent normal tissues.²¹ Chronic oxidative stress may induce DNA damage such as base modifications and strand breaks, which may lead to further mtDNA mutation and chromosomal aberrations that are commonly seen in cancers. Accumulation of oxidative damage and sequence variations in nuclear DNA and mtDNA may ultimately lead to uncontrolled cell growth and malignant transformation of affected tissue cells.

In the present study, a dramatic decrease in the copy number of mtDNA was found in most HCCs (57.4%, FIG. 2) and gastric cancers (54.8%, FIG. 3), but only in 22.6% of lung cancers (FIG. 4), and in 28.0% of colorectal cancers (FIG. 5), respectively. The incidence of the decrease in mtDNA content of these cancers was not increased with their clinical stages. These results indicate that alterations (an increase or decrease) in the copy number of mtDNA occur in the same type of carcinoma, and some tissue-specific factors are involved in reducing the mtDNA copy number of the

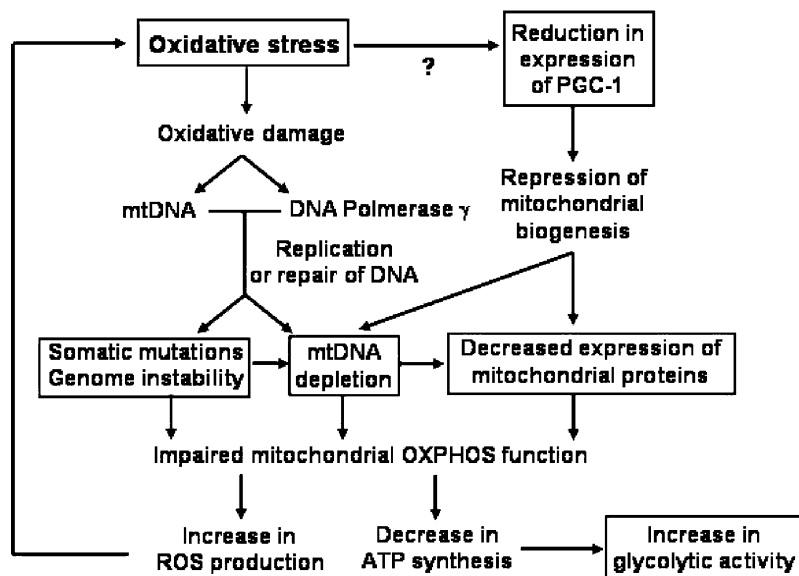


FIGURE 6. Alterations in mitochondria of cancer cells and their possible linkages. Extensive oxidative stress in cancer cells may result in oxidative damages to mitochondrial DNA (mtDNA) and to DNA polymerase γ , which both contribute to somatic mutation and genome instability in mtDNA as well as mtDNA depletion. Moreover, reduced expression of the PPAR γ coactivator-1 (PGC-1) could lead to a repression in mitochondrial biogenesis, thereby causing mtDNA depletion and decreased expression of mitochondrial proteins. Mitochondrial genome instability and mtDNA depletion, as well as mitochondrial biogenesis repression, result in the impaired mitochondrial oxidative phosphorylation (OXPHOS) function, which might cause an increase in the production of reactive oxygen species (ROS) and a decrease in the synthesis of ATP. The increased ROS production enhances oxidative stress, and the decreased ATP synthesis stimulates glycolytic activity in cancer cells. The alterations described in the boxes have been observed in several human cancers.

liver and gastric cancers. However, no dominant factors are involved in the alteration of the mtDNA copy number of the lung or colorectal tissues during carcinogenesis.

During the past decade, a number of factors involved in the replication and maintenance of mtDNA, as well as in biogenesis of mitochondria, have been characterized.²² A transcriptional coactivator, peroxisome proliferator-activated receptor γ coactivator-1 (PGC-1) has been demonstrated to interact with NRF-1 and NRF-2 at the upstream level in the regulation of mitochondrial biogenesis in animal tissues.²³ Recently, the expression of PGC-1 was found to decrease in human HCC, breast, and colon cancers.^{20,24,25} It is possible that carcinogenesis of these tissues could be associated with alterations in the quantity and activity of these regulatory factors. The depletion of mtDNA may be the result of the repression of mitochondrial biogenesis (FIG. 6). The observations are consistent with the findings that there is a general downregulation of mitochondrial components and a repression of the program of mi-

tochondrial proliferation in liver cancers.^{4,20,26} Our results suggest that in most HCCs and gastric cancers, as well as some other carcinomas (lung and colorectal cancers), carcinogenesis is accompanied by an alteration in mitochondrial biogenesis and a repression of mtDNA replication.⁵

In contrast, we observed an increase in the copy number of mtDNA in 48.4% lung cancers, 40.0% colorectal cancers, 27.8% HCCs, and 22.6% gastric cancers. The increase in mtDNA copy number may result from the compensation for the increased energy need and/or the decreased capacity of oxidative phosphorylation in these cancers. It is possible that the increased mitochondrial biogenesis might be similar to the development of ragged-red fibers as a result of overproliferation of mitochondria in muscle of patients suffering from mitochondrial myopathies.^{8,9} Indeed, it was recently reported that the rate of ATP synthesis is decreased in thyroid oncocyoma although the numbers of mitochondria and mtDNA are increased.²⁷ These are consistent with the previous observations in human kidney and colon cancers⁴ that, although there is a selective decrease in the expression of some proteins involved in mitochondrial respiratory function, it may not affect the replication of mtDNA. Thus, the increase in the mtDNA copy number in cancer cells may be a result of the compensation for the mitochondrial dysfunction and energy shortage. Recently, we have demonstrated that oxidative stress can cause an increase in mitochondria and mtDNA content in human cells.²⁸ Further study is necessary to establish the role of the increase of mtDNA in cancer biology. It is possible that elevated ROS production from impaired mitochondrial respiration might be involved in the upregulation of the copy number of mtDNA in carcinomas.

Cancer cells possess mitochondrial alterations that may contribute to resistance to apoptosis and/or cancer development. It has been known since the 1930s that cancerous cells exhibit enhanced glycolytic ATP generation and decreased oxidative phosphorylation, even under normal oxygen tension—a phenomenon known as the Warburg effect.¹ Moreover, it was reported that many cancer cells have a higher mitochondrial transmembrane potential than do the control cells.²⁹ Another common feature of cancer cells is their intrinsic oxidative stress.²¹ In the present study, the observed mutations in the D-loop region and the alterations in the copy number of mtDNA were often seen in many cancers, which may play a central role in mitochondrial alterations in cancer cells and/or in the multistage process of carcinogenesis (FIG. 6). Cancer cells have increased glycolytic activity, perhaps reflecting a less efficient ATP generation in mitochondria due to respiration injury and/or mtDNA mutation as well as depletion, thus forcing the cancer cells to increase glycolysis to produce ATP required for cellular functions (FIG. 6). These metabolic alterations might be linked to a change that contributes to the resistance and compromised apoptotic potential of cancer cells.³⁰ Defects in the execution of apoptosis enable neoplastic cells to survive beyond their intended life span, allowing them time to accumulate genetic alterations that deregulate cell proliferation, increase cell motility, and effect other changes that play an important role in tumor pathogenesis. In addition, a lower coupling efficiency of mitochondrial enzymes may lead to more leakage of electrons from the respiratory enzyme complexes to react with oxygen, forming superoxide anions and other ROS (FIG. 6). The increased oxidative stress in cancer cells may lead to further mutagenesis of both mtDNA and nuclear DNA, activation of proto-oncogenes and some transcription factors, genomic instability, chemotherapy resistance, invasion, and metastasis.²¹ Therefore, it is possible that

alterations in the sequence and copy number of mtDNA in the cancers play some roles in the carcinogenesis and the progression of human cancers as well as resistance to chemotherapy.

In conclusion, we have shown differential alterations in the copy number and mutations of mtDNA that occur in human liver, gastric, lung, and colorectal cancers. Our findings suggest that mtDNA instability, together with the alteration in the copy number of mtDNA, plays an important role in the carcinogenesis of human cancers. The molecular mechanism by which the mtDNA copy number is changed by cancer-associated mutations in the D-loop region awaits further investigation.

ACKNOWLEDGMENTS

This work was supported by grants TCVGH-917106D, TCVGH-927107D, TCVGH-93106D from Taichung Veterans General Hospital, and also by grants NSC 91-2320-B-040-040, NSC 91-2320-B-010-069, NSC 92-2320-B-010-077, NSC 91-2745-B-040-001, NSC 92-2320-B-010-037, and NSC 92-2745-B-040-001 from the National Science Council, Taiwan, Republic of China.

REFERENCES

1. WARBURG, O. 1930. *Metabolism of Tumors*. Arnold Constable. London.
2. DANG, C.V. & G.L. SEMENZA. 1999. Oncogenic alterations of metabolism. *Trends Biochem. Sci.* **24**: 68–72.
3. LEE, H.C. & Y.H. WEI. 2000. Mitochondrial role in life and death of the cell. *J. Biomed. Sci.* **7**: 215.
4. CUEZVA, J.M., M. KRAJEWSKA, M.L. DE HEREDIA, *et al.* 2002. The bioenergetic signature of cancer: a marker of tumor progression. *Cancer Res.* **62**: 6674–6681.
5. ISIDORO, A., M. MARTINEZ, P.L. FERNANDEZ, *et al.* 2004. Alteration of the bioenergetic phenotype of mitochondria is a hallmark of breast, gastric, lung and oesophageal cancer. *Biochem. J.* **378**: 17–20.
6. ATTARDI, G. & G. SCHATZ. 1988. Biogenesis of mitochondria. *Annu. Rev. Cell Biol.* **4**: 289–333.
7. CROTEAU, D.L. & V.A. BOHR. 1997. Repair of oxidative damage to nuclear and mitochondrial DNA in mammalian cells. *J. Biol. Chem.* **272**: 25409–25412.
8. WALLACE, D.C. 1999. Mitochondrial diseases in man and mouse. *Science* **283**: 1482–1488.
9. WEI, Y.H. & H.C. LEE. 2003. Mitochondrial DNA mutations and oxidative stress in mitochondrial diseases. *Adv. Clin. Chem.* **37**: 83–128.
10. POLYAK, K., Y. LI, H. ZHU, *et al.* 1998. Somatic mutations of the mitochondrial genome in human colorectal tumors. *Nat. Genet.* **20**: 291–293.
11. FLISS, M.S., H. USADEL, O.L. CABALLERO, *et al.* 2000. Facile detection of mitochondrial DNA mutations in tumors and bodily fluids. *Science* **287**: 2017–2019.
12. PENTA, J.S., F.M. JOHNSON, J.T. WACHSMAN, *et al.* 2001. Mitochondrial DNA in human malignancy. *Mutat. Res.* **488**: 119–133.
13. SANCHEZ-CEPEDES, M., P. PARRELLA, S. NOMOTO, *et al.* 2001. Identification of a mononucleotide repeat as a major target for mitochondrial DNA alterations in human tumor. *Cancer Res.* **61**: 7015–7019.
14. NOMOTO, S., K. YAMASHITA, K. KOSHIKAWA, *et al.* 2002. Mitochondrial D-loop mutations as clonal markers in multicentric hepatocellular carcinoma and plasma. *Clin. Cancer Res.* **8**: 481–487.
15. SHADEL, G.S. & D.A. CLAYTON. 1997. Mitochondrial DNA maintenance in vertebrates. *Annu. Rev. Biochem.* **66**: 409–435.

16. LEE, H.C., C.Y. LU, H.J. FAHN, *et al.* 1998. Aging- and smoking-associated alteration in the relative content of mitochondrial DNA in human lung. *FEBS Lett.* **441**: 292–296.
17. LEE, H.C., S.H. LI, J.C. LIN, *et al.* 2004. Somatic mutations in the D-loop and decrease in the copy number of mitochondrial DNA in human hepatocellular carcinoma. *Mutat. Res.* **547**: 71–78.
18. MAMBO, E., X. GAO, Y. COHEN, *et al.* 2003. Electrophile and oxidant damage of mitochondrial DNA leading to rapid evolution of homoplasmic mutations. *Proc. Natl. Acad. Sci. USA* **100**: 1838–1843.
19. GRAZIEWICZ, M.A., B.J. DAY & W.C. COPELAND. 2002. The mitochondrial DNA polymerase as a target of oxidative damage. *Nucleic Acids Res.* **30**: 2817–2824.
20. YIN, P.H., H.C. LEE, G.Y. CHAU, *et al.* 2004. Alteration of the copy number and deletion of mitochondrial DNA in human hepatocellular carcinoma. *Br. J. Cancer* **90**: 2390–2396.
21. TOYOKUNI, S., K. OKAMOTO, J. YODOI, *et al.* 1995. Persistent oxidative stress in cancer. *FEBS Lett.* **358**: 1–3.
22. MORAES, C.T. 2001. What regulates mitochondrial DNA copy number in animal cells? *Trends Genet.* **17**: 199–205.
23. KNUTTI, D. & A. KRALLI. 2001. PGC-1, a versatile coactivator. *Trends Endocrinol. Metab.* **12**: 360–365.
24. JIANG, W.G., A. DOUGLAS-JONES & R.E. MANSEL. 2003. Expression of peroxisome-proliferator activated receptor-gamma (PPAR γ) and the PPAR γ co-activator, PGC-1, in human breast cancer correlates with clinical outcomes. *Int. J. Cancer* **106**: 752–757.
25. FEILCHENFELDT, J., M.A. BRUNDLER, C. SORAVIA, *et al.* 2004. Peroxisome proliferators-activated receptors (PPARs) and associated transcription factors in colon cancer: reduced expression of PPAR γ -coactivator 1 (PGC-1). *Cancer Lett.* **203**: 25–33.
26. CUEZVA, J.M., L.K. OSTRONOFF, J. RICART, *et al.* 1997. Mitochondrial biogenesis in the liver during development and oncogenesis. *J. Bioenerg. Biomembr.* **29**: 365–377.
27. SAVAGNER, F., B. FRANC, S. GUYETANT, *et al.* 2001. Defective mitochondrial ATP synthesis in oxyphilic thyroid tumors. *J. Clin. Endocrinol. Metab.* **86**: 4920–4925.
28. LEE, H.C., P.H. YIN, C.Y. LU, *et al.* 2000. Increase of mitochondria and mitochondrial DNA in response to oxidative stress in human cells. *Biochem. J.* **348**: 425–432.
29. CHEN, L.B. 1988. Mitochondrial membrane potential in living cells. *Annu. Rev. Cell Biol.* **4**: 155–181.
30. CAVALLI, L.R. & B.C. LIANG. 1998. Mutagenesis, tumorigenicity, and apoptosis: are the mitochondria involved? *Mutat. Res.* **398**: 12–26.